Solution Requirements

Date	01 October 2026
Team ID	SWTID-2026-1940
Project Name	Team Enrollment Confirmation - EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what the EduGenie solution will do and how well it will perform.

• **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (what the system should do).
• **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of the system, rather than specific behaviors (how the system should be).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	**Authentication**	Students/users shall register and securely log in to EduGenie using valid credentials. The system shall validate credentials before granting access.	**High**
2	**Authorization levels**	Students can view their profile, enroll in learning programs and use the Gemini learning assistant. Admins can manage courses, enrollment records and reports.	**High**
3	**External interfaces**	EduGenie shall connect to the Google Gemini API to send learning prompts and receive AI-generated educational responses. It may also use email/notification services for confirmation.	**High**
4	**Transactions processing**	The system shall capture enrollment requests, validate the selected course/program, store the enrollment record and generate a confirmation for the student.	**High**

Solution Requirements — Functional Requirements

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
5	**Reporting**	The system shall provide enrollment status, confirmation history, learning activity summaries and basic administrative reports.	**Medium**
6	**Business rules, etc.**	A student shall not receive duplicate enrollment confirmations for the same active enrollment. Required fields must be completed before enrollment is submitted. Enrollment status shall be tracked as Pending, Confirmed or Cancelled.	**High**
7	**Compliance to laws or regulations**	The system shall protect personal information, collect only necessary student data and provide appropriate privacy controls. API usage shall follow the applicable service terms and privacy requirements.	**High**
8	**Other**	The system shall display clear confirmation messages, allow users to view their enrollment details and provide an accessible interface for learning assistance.	**Medium**

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	**Performance & Speed**	The application should load the main pages quickly and process normal enrollment requests without noticeable delay.	**Target:** Typical page response within 2 seconds under normal load.
2	**Scalability**	The system should support growth in students, courses, enrollment records and AI learning requests without major redesign.	**Target:** Support at least 10,000 registered users with scalable storage.
3	**Security & Data Privacy**	Protect credentials and student information using secure authentication, access control, encrypted connections and safe API-key handling.	**Target:** HTTPS/TLS, role-based access and no exposed API keys.
4	**Reliability & Availability**	The application should handle temporary failures gracefully and avoid loss or duplication of enrollment records.	**Target:** 99% monthly availability and reliable record persistence.
5	**Usability & Accessibility**	The interface should be simple for students and admins, with readable text, clear navigation, understandable messages and keyboard-friendly controls where applicable.	**Target:** Core enrollment flow completed without assistance by a first-time user.
6	**Other**	The solution should be maintainable, modular and documented so team members can update features and troubleshoot issues.	**Target:** Modular components, basic technical documentation and test checklist.

Team Work Assignment

Work is assigned according to the exact order of names provided.

S.No	Team Member	Assigned Work
1	**R.Priya dharshini**	Project coordination; requirement gathering; Functional Requirements (FR) 1–2; authentication and authorization flow; final team coordination.
2	**S.Shobana**	FR 3–4; Google Gemini API/external interface planning; enrollment transaction processing; database and enrollment data flow support.
3	**S.Rakshana**	FR 5–6; reporting requirements; business rules; enrollment-status logic; testing of confirmation and duplicate-enrollment handling.
4	**E.Shylaja**	FR 7–8; privacy/compliance requirements; usability requirements; student confirmation and notification flow; user-interface review.
5	**S.Sheeba**	NFR 1–6; performance, scalability, security, reliability, accessibility and maintainability requirements; documentation and final quality review.

Project Summary

EduGenie is a Google Gemini powered learning assistant designed to support students with learning assistance while also managing team/student enrollment confirmation. The system combines secure user access, enrollment processing, confirmation notifications, data storage and AI-powered learning support.

Key Acceptance Points

1	A student can register/login and access the EduGenie system.
2	A student can submit an enrollment request and receive a confirmation/status.
3	The system stores enrollment and learning-history data correctly.
4	The Google Gemini integration provides learning assistance through controlled API requests.
5	Admin functions, privacy controls, security and basic performance requirements are addressed.